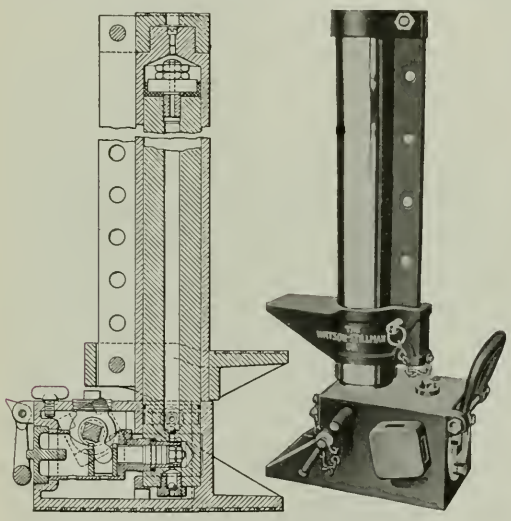
## EMERGENCY JACK

A hydraulic jack, the design of which embodies a number of unusual features, has recently been developed by the Watson-Stillman Company, New York. This jack was designed primarily to meet the demands of an emergency jack for street railway use, but its flexibility of adjustment is such as to make it of value for a variety of purposes wherever lifting work is performed.

The construction of the jack is shown in the sectional elevations, from which it will be seen that the cylinder is the moving part instead of the ram as in the usual type of jack. This allows the pump mechanism to stay in a fixed vertical position and permits the working parts of the jack to be made simpler and more compact than is usually the case. The piston is packed with leather rings and the valves are of the ball type with all passages amply proportioned. The pressure is relieved by means of a key operating a small needle valve. The jack is operated with a special oil, which not only acts as a lubricant, but prevents rust on the working parts and the possibility of freezing. It has no detrimental effect on the packings.

One of the most notable features of this jack is the arrangement of the claw which can be moved vertically upon the cylinder and adjusted to the most convenient height. The claw and the cylinder can be swung through a complete circle without changing the position of the jack or the location of the pump. The operating lever is but 18 in. long, but one man weighing 125 lb. can obtain the maximum pressure with but slight effort. The lever is curved and the socket has a hole in each of its four sides to allow for convenience in operation from practically any position. The jacks are now built in five and ten ton sizes with a ram stroke of 10 in., and are guaranteed by the manufacturer to stand a 50 per cent overload without detriment to any of the parts.



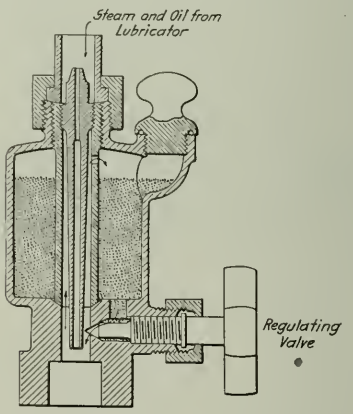
Exterior and Sectional Views of Emergency Jack

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Timber Consumption of an Hawaiian Railway.—The Hilo Railroad, operating about 72 miles of line on the island of Hawaii and the only standard gage railway in the territory of Hawaii uses about 1,000,000 feet of lumber yearly. The company has car shops and a foundry, and builds and repairs its own cars at Hilo. Redwood ties are used in the roadbed. Douglas fir is chiefly used in the shops, though ohia has been found well adapted for car roofs.

## GRAPHITE LUBRICATOR

The Graphite Lubricator Company, Detroit, Mich., has placed on the market a graphite lubricator for lubricating with flake graphite the valves and cylinders of locomotives. This graphite is suspended in common engine oil for the purpose of more easily filling the lubricator. The device is used in connection with the oil lubricator as now in use on locomotives. The present oil lubricator acts as the cleaner, while the graphite lubricator delivers graphite to the cylinders and valves as the lubricant. The accompanying illustration clearly shows its construction. The oil line from the steam or gravity-feed lubricator in the cab is connected to the top of the graphite lubricator, and the oil and steam pass to the cylinder and valves through the nozzle shown on the inside. The graphite is contained in the annular space surrounding this nozzle, and the feed is controlled by means of the regulating valve, as shown. The pulsating pressure of the steam will cause the feeding of the graphite through the passage controlled by the regulating valve, and there it becomes mixed with the oil from the gravity-feed lubricator. It is then distributed to the various parts of the valves and cylinders. This lubricator is adapted for use on superheater and compound locomotives (either piston or slide valve type) and especially those engines which experience trouble in obtaining the proper lubrication of the valves and cylinders. It is claimed that the priming of locomotives will not in any way interfere with the proper lubrication of the cylinders and valves where this lubricator is used. It is also claimed that 50 per cent less cylinder oil will be required for lubrication in the cylinders and valves, and that at the same time a better lubrication will be obtained by the use of the flake graphite.




Cylinder Lubricator for Flake Graphite

The lubricator shown, which will hold three ounces of the graphite lubricating mixture, is filled through the plug shown at the top and to the right of the lubricator. It may be attached to any standard steam chest choke-plug fitting, and thus eliminates the use of the choke plugs. One filling of the model A three-ounce lubricator is sufficient for the operation of a locomotive for approximately 250 miles without further attention on the part of the crew. At present the lubricators are made in two sizes—model A, which holds three ounces of lubricant, and model B, which holds six ounces.

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German Locomotives Dismantled for Copper.—Unconfirmed press despatches from Amsterdam report that the scarcity of copper is such throughout Germany that orders have been given to dismantle as many locomotives on the state railways as possible, in order to use the copper and brass for making ammunition.